Week 2 · Rules and decisions

Programming for Information Systems

Who chooses the rule, and how does a program apply it?

Class warm-up · Research 1 exchange

In groups of three, share the information system you researched.

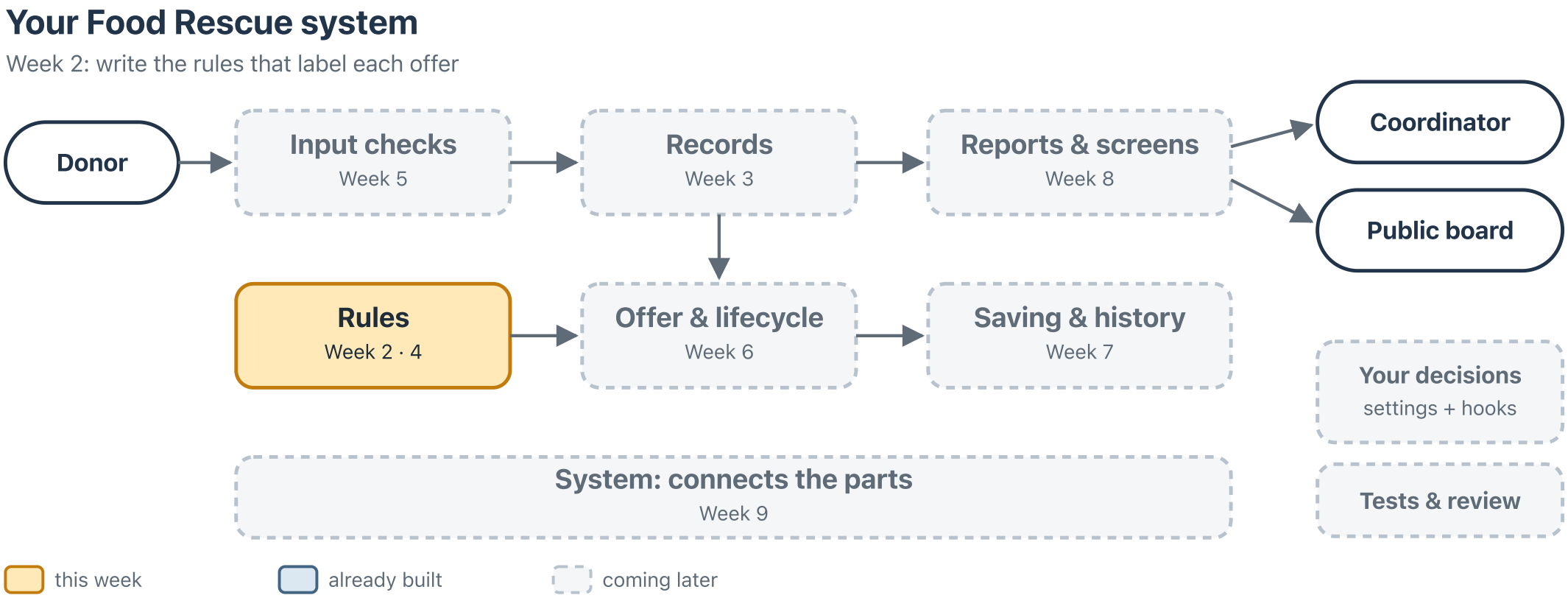
1. Who uses it, and what data does it keep?
2. What **rule** did you find? Is it written down, or only in people's heads?
3. What could go wrong, and who would be hurt?

> 📝 Write down one idea from a classmate that you want to remember.

This week

Question	How does a system make a decision, and who chooses the rule?
Python	comparisons, `True` / `False`, `and` / `or` / `not`, `if` / `elif` / `else`, a first small function
Principle 2	Explicit rules: name your numbers, cover every case, test the boundary
You build	`pickup_label` and `urgency`

Your system, by Week 10



The idea: business rules

A **business rule** is a decision about how work is done.

- "*An exam mark of 50 or more is a pass.*"
- "*Children under 6 travel free.*"

> **Coordinator:** "Not approved yet → **hold**. Approved and **20 portions or more** → **team**. Otherwise → **standard**."

- People choose the rule. The program applies it the same way, every time.
- A wrong rule is applied wrongly every time, so rules must be **written exactly** and **tested at the edges**.

Python foundations

How a program decides

Comparisons give `True` or `False`

```
portions = 20
print(portions >= 20)       # True
print(portions > 20)        # False
print(portions == 20)       # True
print("bread" == "Bread")   # False
```

- `==` `!=` `<` `<=` `>` `>=`
- "20 or more" is `>= 20`. "More than 20" is `> 20`.
- Text comparison is exact: capital letters matter.

Combining conditions: `and`, `or`, `not`

```
approved = True
portions = 25
print(approved and portions >= 20)       # both must be True
print(not approved)                      # the opposite
print(portions >= 20 or portions < 0)    # at least one True
```

- `and`: every part must be `True`.
- `or`: at least one part must be `True`.
- `not`: turns `True` into `False`, and the other way.

`if` and `else`: choosing what to run

```
temperature = 31
if temperature >= 30:
    print("Hot: keep the food in the shade")
else:
    print("Normal temperature")
print("This line always runs")
```

- The `if` line ends with a colon `:`.
- The lines that belong to it are **indented** by 4 spaces.
- Indentation is not decoration: it decides what belongs to what.

`elif`: checked in order, first match wins

```
mark = 72
if mark >= 80:
    grade = "A"
elif mark >= 65:
    grade = "B"
elif mark >= 50:
    grade = "C"
else:
    grade = "fail"
print(grade)
```

🤔 What grade does exactly 65 get? And 64?

Name your numbers

```
TEAM_PICKUP_MIN_PORTIONS = 20      # decided by the coordinator

if portions >= TEAM_PICKUP_MIN_PORTIONS:
    ...
```

- A number inside a rule is a **decision**.
- A name lets people **find** it, **understand** it and **change** it in one place.
- CAPITAL names are **constants**: the program does not change them while it runs.

A first function: package a rule

```
def is_hot(temperature):
    if temperature >= 30:
        return True
    return False

print(is_hot(31))    # True
print(is_hot(25))    # False
```

- `def` starts the function; `temperature` is its input.
- `return` sends the answer back and **stops** the function.
- The system calls the same rule for every offer.

Test the boundary

```
def needs_team(portions):
    if portions > 20:       # rule: "20 or more"
        return True
    return False

print(19, needs_team(19))   # False  ✓
print(20, needs_team(20))   # False  ✗  should be True
print(21, needs_team(21))   # True   ✓
```

Only the **boundary** case shows the bug. Always test one below, exactly at, and one above.

⚖️ Good or risky?

Version A (from an AI assistant)

```
def label(p, a):
    if p > 20 and a == True:
        return "team"
    elif a == True:
        return "standard"
```

Version B

```
TEAM_PICKUP_MIN_PORTIONS = 20

def pickup_label(portions, approved):
    if not approved:
        return "hold"
    if portions >= TEAM_PICKUP_MIN_PORTIONS:
        return "team"
    return "standard"
```

Principle 2 · Explicit rules

- **Name your numbers**: a constant shows the decision.
- **Cover every case**: every path ends with a `return`.
- **Test the boundary**: one below, exactly at, one above.

> When you review AI code, ask: *what happens exactly at the limit, and when the condition is False?*

You build: the two pickup rules

```
TEAM_PICKUP_MIN_PORTIONS = 20

def pickup_label(portions, approved):
    # not approved → "hold"; at least the minimum → "team"
    # otherwise → "standard"
    return "?"

fr.check("pickup_label", pickup_label,
         {"TEAM_PICKUP_MIN_PORTIONS": TEAM_PICKUP_MIN_PORTIONS})
```

- `urgency(hours_left)`: ≤ `URGENT_HOURS` → "urgent", ≤ `SOON_HOURS` → "soon", else "later".
- The check tests the boundaries **of your own numbers**.

Plug your rules into the real system

```
MY_SETTINGS = {"TEAM_PICKUP_MIN_PORTIONS": TEAM_PICKUP_MIN_PORTIONS,
               "URGENT_HOURS": URGENT_HOURS, "SOON_HOURS": SOON_HOURS}
my_system = fr.System(MY_SETTINGS,
                      hooks={"pickup_label": pickup_label, "urgency": urgency})
my_system.run_day(show=False)
print(my_system.report())
```

- A **hook** is a place where the system lets you plug in your own function.
- If your function crashes, the system prints ⚠️ and uses its default rule.

🧭 Your decision · team size and urgency

- Change `TEAM_PICKUP_MIN_PORTIONS`, `URGENT_HOURS`, `SOON_HOURS` (or keep them and explain why).
- Rerun: the checks still pass, and the labels change.

Decision log	
My choice	team from … portions; urgent ≤ … h; soon ≤ … h
An alternative	…
Why	…
Who is affected	…

This week's tasks

- ⭐ **Core:** P1–P4, three problems in Version A, `pickup_label` and `urgency` with all ✅, plugged in, decision log.
- ⭐⭐ **Stretch:** S1 `send_team` with two conditions; `explain_label` that says why.
- 🔎 **Research 2 (5%):** a real-life rule written exactly, with a normal, a boundary and an odd case. Share it in the Week 3 warm-up.

Next week · Records and data

One offer is easy. A whole day has many. You will keep **one record per offer**, add them up, and find any offer by its **ID**.